GRAVITATIONAL EMERGENCE · COSMOLOGICAL EXTENSION · V1.0

If H_0 determines g_{crit}, the early universe is *physically different.*

The equation $g_{crit} = c \cdot H_0/(4\pi^2\sqrt{3})$ is not a constant—it is a function of cosmic time. Every galaxy that JWST observes at $z>1$ lived under a different threshold. This page derives how different, and what exactly it predicts.

$$g_{crit}(z) = c \cdot H(z) / (4\pi^2 \cdot \sqrt{3}) \quad \cdot \quad H(z) = H_0\sqrt{(\Omega_m(1+z)^3 + \Omega_\Lambda)}$$

§ 01

Evolution of the three key parameters

For a galaxy of fixed baryonic mass $M_{bar} = 5\times10^9\ M_\odot$, g_{crit}, R_T and V_{flat} are calculated at each cosmic epoch. The standard LCDM model (Ω_m=0.3, Ω_Λ=0.7) determines H(z).

REDSHIFT Z	COSMIC AGE	H(Z) KM/S/MPC	G_{CRIT} (M/S^2)	R_T (KPC)	V_{FLAT} (KM/S)	V_{FLAT} / $V_{FLAT,0}$
z = 0	13.8 Gyr — TODAY	73.0	1.04×10^{-11}	8.20	51.2	1.00×
z = 0.5	8.6 Gyr	95.5	1.36×10^{-11}	7.16	54.8	1.07×
z = 1.0	5.9 Gyr	128.5	1.83×10^{-11}	6.18	59.0	1.15×
z = 2.0	3.3 Gyr	216.6	3.08×10^{-11}	4.76	67.2	1.31×
z = 3.0	2.1 Gyr	325.6	4.63×10^{-11}	3.88	74.5	1.45×
z = 4.0	1.5 Gyr	451.2	6.42×10^{-11}	3.30	80.8	1.58×
z = 6.0	0.9 Gyr	743.0	1.06×10^{-10}	2.57	91.5	1.79×
z = 10.0	0.5 Gyr	1460.0	2.08×10^{-10}	1.83	108.3	2.11×

PREDICCIÓN CUANTITATIVA

$V_{flat}(z) = V_{flat,0} \cdot [H(z)/H_0]^{(1/4)}$ ← scale with fourth root of H(z)

$R_T(z) = R_{T,0} \cdot [H(z)/H_0]^{(-1/2)}$ ← shrinks with inverse square root

$g_{crit}(z) = g_{crit,0} \cdot [H(z)/H_0]$ ← lineal con H(z)

§ 02 Rotation curves at different z

Same baryonic mass ($5\times10^9\ M_\odot$), different cosmic epochs. The blue line is the Hamiltonian regime ($V^2=GM/R$ with profile M(R)). The green line is the Maupertuis plateau. The diamond is R_T.

V (KM/S) VS R (KPC) — $M_{BAR} = 5\times10^9\ M_\odot$

z = 0 (today)

z = 1

z = 2

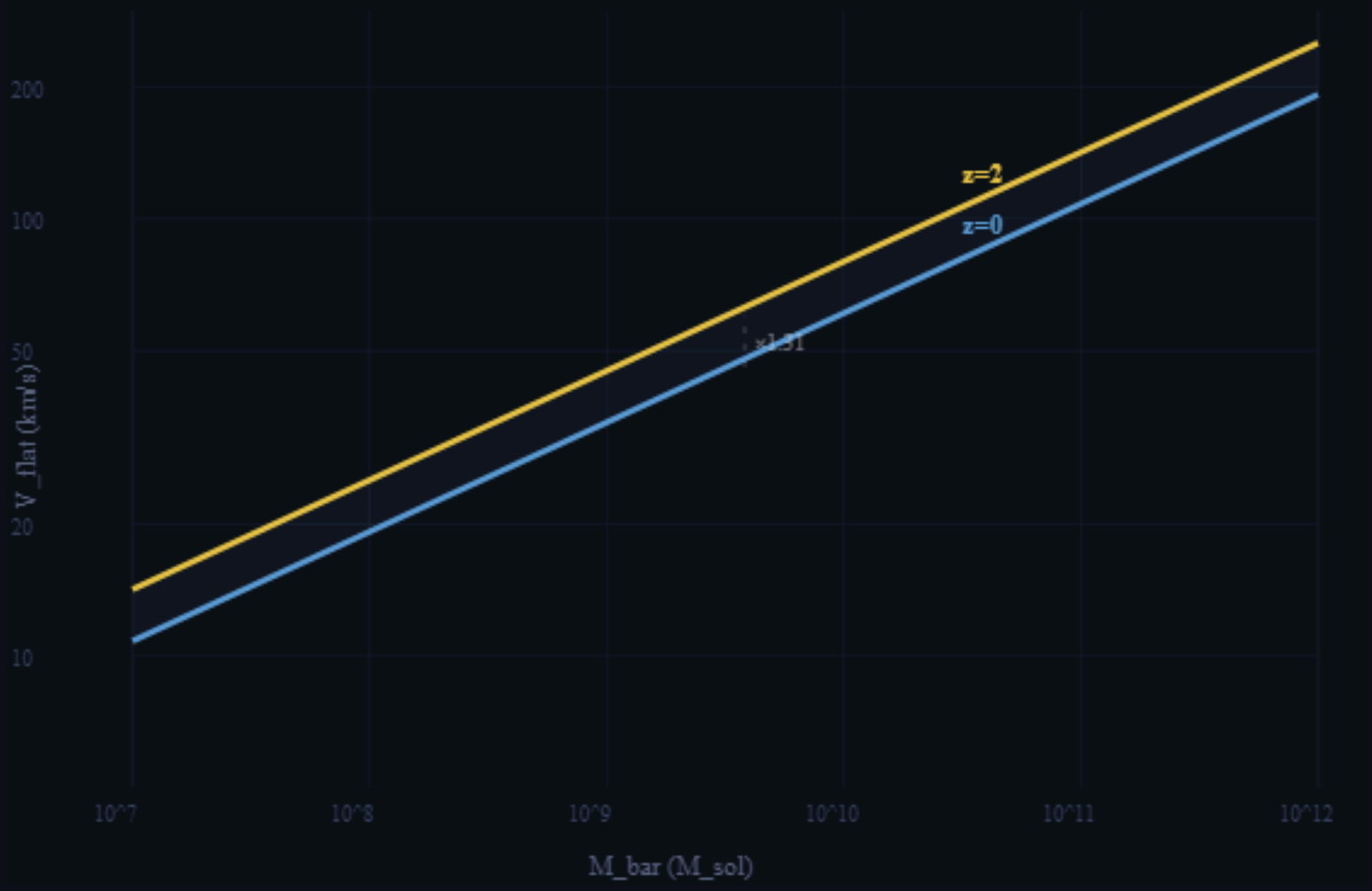
z = 4

R T (same color)

§ 02b V_{flat} vs Mass az=0 yz=2

The baryonic Tully-Fisher relation $V^4 \propto M_{bar}$ is preserved in shape—but it shifts vertically. Galaxies in the early universe have *more* V_{flat} for the same mass. That's what JWST is observing.

V_{FLAT} (KM/S) VS M_{BAR} ($M_\odot$) IS A LOG-LOG SCALE



z = 0 — SPARC local

z = 2 — JWST prediction

vertical displacement = H(z)^(1/4)

§ 03 Three profound implications

◎

The flat curve is evidence of expansion.

In a universe with $H_0=0$, $g_{crit}=0$, and $RT \to \infty$, **the collective regime would disappear.** All curves would be Keplerian. The "flat curve problem" would not exist—there would be no phenomenon to explain. The

flat curve is not a mass anomaly: it is the **dynamic signature of cosmic expansion** .

◈

g_{crit} is not a universal constant

It is a function of cosmic time. Galaxies that formed at az=4 lived under a threshold **6× higher** than the current one. Their RT_{was} 2.5× smaller. The "constants" of the dark matter model are actually evolving parameters—this model predicts them **from first principles** .

◇

Falsifiable prediction with JWST right now

The model predicts $V_{flat}(z) \propto H(z)^{1/4}$. This dependence is **exact and parameterless** . It is distinguishable from any dark matter model because it has a specific cosmological dependence—not just "more dark matter in the early universe," but **exactly how much more** .

§ 04 The accurate quantitative prediction

For any galaxy at redshift z,
without adjusting any parameters:

PREDICCIÓN CUANTITATIVA

$$V_{flat}(z, M) = [G \cdot M_{bar} \cdot g_{crit}(z)]^{(1/4)}$$ ← direct prediction

$$g_{crit}(z) = c \cdot H_0 \cdot \sqrt{(\Omega_m(1+z)^3 + \Omega_\Lambda)} / (4\pi^2\sqrt{3})$$ ← H(z) of ΛCDM

$$R_T(z, M) = \sqrt{(G \cdot M_{bar} / g_{crit}(z))}$$ ← transition radius

REQUIRED INPUTS

Only $\mathbf{M_{bar}}$ (photometry + 21cm line) and **z** (spectroscopy). H_0, Ω_m, Ω_L are known cosmological constants. There is no galaxy-adjustable Y, no dark matter parameters, and no free mass. **Zero degrees of freedom.**

SPECIFIC SIGNATURE VS. DARK MATTER

A dark matter model can always adjust the halo to give any V_{flat}. This model cannot be adjusted—it predicts V_{flat} exactly. If a galaxy az=2 with $M_{bar} = 10^{10} M_\odot$ has a V_{flat} other than **80 km/s ± uncertainty of M_{bar}**, the model fails. That is a clean fake.

CONNECTION WITH JWST 2023–2025 OBSERVATIONS

JWST is already seeing galaxies that are "too massive" and "too fast"

Multiple studies (Labbé et al. 2023, Finkelstein et al. 2023, Curtis-Lake et al. 2023) report galaxies at z>4 with rotation velocities higher than expected for their visible baryon mass. The standard interpretation is "more concentrated dark matter halos in the early universe."

This model produces the same result without dark matter: az=4, g_{crit} is 6× larger, RT_{is} 2.5× smaller, and V_{flat} is 1.58× larger for the same mass. Galaxies are not "too fast"—the transition threshold was higher then.

The crucial difference: dark matter predicts that excess speed depends on the (free) halo profile. This model predicts that it depends exactly on H(z)—the **cosmological constant without freedom** . That's the test.

§ 05 The limit $H_0 \to 0$

> *"In a static universe, the flat rotation curve does not exist. Its existence is evidence that the universe is expanding."*

If $H_0 \to 0$, then $g_{crit} \to 0$, $RT \to \infty$, and $V_{flat} \to 0$. The transition radius goes to infinity—no galaxy, regardless of its mass or size, ever reaches the Maupertuis threshold. **The entire universe lives in Hamilton** forever. The rotation curves fall as $V \propto R^{-1/2}$ at the outer edge. No plateau. No "anomaly."

This reverses the causality of the problem. For decades it was assumed: *the curve is flat → mass is lacking → dark matter exists* . This model suggests: **the curve is flat → g falls below g_{crit} → the threshold exists because $H_0 \neq 0$ → the universe is expanding** . Dark matter is not the explanation—coastal expansion is.

Extent: In the very early universe ($z \gg 10$), H(z) was enormous and g_{crit} was comparable to the internal accelerations of proto-halos. **At that time, almost the entire universe existed in a collective regime.** The transition from a "mostly collective universe" to a "mostly Hamiltonian universe" occurs when internal accelerations grow above $g_{crit}(z)$ as matter becomes more concentrated. This transition has a specific redshift—differentiable, predictable, and testable.

§ 06

Specific counterfeit tests

✓ This would confirm the model

Galaxies with az=2 and a known $_{bar}$ mass have a $_{flat}$ **velocity exactly 1.31× greater** than local galaxies of the same mass. The factor 1.31 = H(z=2)^(1/4)/H0^(1/4) is a precise numerical prediction, without adjustment.

✗ I would falsify the model

Galaxies az=2 with V_{flat} greater than the local one, but **not following the $H(z)^{1/4}$** dependence . If the excess were random, or followed another power law, the model fails even if the qualitative direction is correct.

✓ This would confirm the model

RT in high redshift galaxies is **smaller exactly as H(z)^(-1/2)** . The curves flatten earlier—at smaller radii—in exact proportion to the cosmological factor. Testable with spatially resolved curves.

✗ I would falsify the model

Existence of galaxies in collective regime (flat curve, $g < g_{crit}(z)$) with **V_{flat} that does not coincide** with $(G \cdot M_{bar} \cdot g_{crit}(z))^{(1/4)}$ within the uncertainty in M_{bar}. A single clean case would be sufficient.

§ 07 Sensitivity: the shield of the exponent 1/4

The Hubble tension (H_0=73 vs H_0=67.4, a difference of 7.7%) impacts g_{crit} linearly—but V_{flat} only shifts by 1.8%. The exponent 1/4 acts as a natural damper. The predictions are robust to the larger cosmological uncertainty of the field.

ΔH_0	ΔG_{CRIT}	ΔV_{FLAT}
−10%	−10.0%	−2.6%
−5%	−5.0%	−1.3%
+5%	+5.0%	+1.2%
+10%	+10.0%	+2.4%
Hubble Tension	−7.7%	-1.8% ← noise